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Polymer solubility in ionic liquids: dominated by hydrogen bonding†

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Polymer solubility in ionic liquids (ILs) cannot be predicted by the solubility parameter approach based on the "like dissolves like" principle. According to the Kamlet–Abraham–Taft (KAT) multi-parameter polarity scale, ILs can be categorized on the basis of hydrogen-bond acidity or basicity ones. The experimental observations, that acidic ILs easily dissolve basic polymers and basic ILs dissolve acidic polymers, reflect the complementary nature of hydrogen-bonding interactions. A quantitative hydrogen-bonding analysis is proposed for predicting the solubility by taking the product of $\Delta\alpha\Delta\beta$ as an indicator of the competition between cross-association and self-association hydrogen bonding (H-bonding), where $\Delta\alpha$ is the difference of acidity parameters between the polymer and IL, and $\Delta\beta$ is the difference of basicity. This solubility criterion has been validated by the solubility data of 19 polymers (11 acidic and 8 basic) in 11 ILs (7 acidic and 4 basic). These principles based on KAT parameters can be applied to other systems dominated by hydrogen bonding.

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Introduction

Ionic liquids (ILs), which have unique properties, such as high ionic conductivity, thermal and (electro)chemical stability, non-volatility, and nonflammability,[1–3] are useful in numerous chemical applications. One of the particular advantages is that they are capable of dissolving complex macromolecules and polymeric materials with high efficiency,[4–6] and then a further chemical transformation or material processing step can be carried out. Moreover, novel materials have been developed by combining ILs and polymers.[7–9] Therefore, it is exceptionally important to understand the solubility of polymers in ILs.[6] ILs have similar polarity,[10] but polymer dissolution in ILs is complex and not readily predicted due to multiple solute–solvent interactions. The solubility parameter (SP, or Hildebrand parameter) δ, based on the principle of "like dissolves like",[11–13] has frequently been applied to account for the solubility of polymers in organic solvents:

$$\begin{aligned}\Delta G_m &= \Delta H_m - T\Delta S_m \\ &= RT\left(\frac{\phi_p}{N_p}\ln\phi_p + \phi_s\ln\phi_s + \chi\phi_p\phi_s\right),\end{aligned} \quad (1)$$

where ΔG_m, ΔH_m, T, and ΔS_m are the Gibbs free energy of mixing, enthalpy of mixing, absolute temperature, and entropy of mixing, respectively. R, ϕ, and N_p are the molar gas constant, volume fraction, degree of polymerization of the polymer, respetcively. χ, which represents the so-called Flory–Huggins interaction parameter, is calculated using the Hildebrand solubility parameter, written as:

$$\chi \sim \frac{(\delta_p - \delta_s)^2}{RT}, \quad (2)$$

and the subscripts p and s refer to the polymer and solvent, respectively.

However, the "like dissolves like" principle cannot serve as a qualitative guide for predicting polymer solubility in ILs as reported in two review papers.[5,6] The solubility parameter method works very well for poly(methyl methacrylate)(PMMA)/organic solvent systems (see Fig. 3 of ref. 14), but it does not work for PMMA/IL systems (see Fig. 5 of ref. 14). For instance, the solubilities of poly(methyl methacrylate) (PMMA, SP = 18.3 $MPa^{1/2}$) are quite different in three ILs ([C_4mim][Tf_2N], [C_6mim][PF_6] and [C_8mim][BF_4]) that have essentially identical solubility parameters (21.5–22.0 $MPa^{1/2}$).[14] On the other hand, poly(dimethyl acrylamide) (PDMAm, 37.0 $MPa^{1/2}$), poly(vinyl pyrrolidone) (PVP, 30.1 $MPa^{1/2}$) and PMMA have good solubilities in [C_4mim][Tf_2N], but polystyrene (PS), which has a similar SP value (18.6 $MPa^{1/2}$) like PMMA, is incompatible with [C_4mim][Tf_2N].[6] Clearly, a single-parameter-polarity-approach is not sufficient to explain the variation of experimental results in numerous IL-mediated processes.[10,15]

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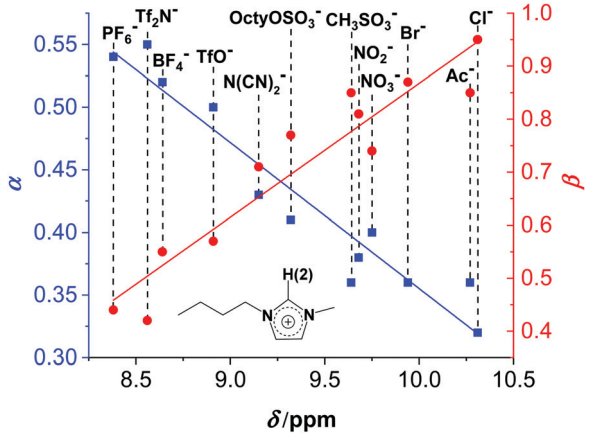


Fig. 1 Correlation of α (squares) and β (circles) with the ^{1}HNMR chemical shift of H2 of [C_4mim]-type ILs (plotted according to data in Table 4 of ref. 20).

The most widely applied method to describe multiple polarity properties is the Kamlet–Abraham–Taft (KAT) polarity scale,[16,17] which has been used extensively for characterizing solvents, solutes and solvent–solute interactions. The KAT polarity properties, evaluated by using different solvatochromic probes,[18,19] are expressed by three independent parameters: π^* is the dipolarity/polarisability, α is the hydrogen bond donating (HBD) or acidity, and β is the hydrogen bond accepting (HBA) or basicity. The physical significance of KAT parameters of ILs has been discussed intensively by Spange *et al.*[20–22] They found correlations between the α and β values of [C_4mim]-type ILs with different anions and ^{1}H NMR chemical shift of H2 of imidazoles (Fig. 1), which also implies the strong connection between the α value of the cation and the β value of the anion in ILs. Therefore, ILs located in the left side may be called as the ILs with hydrogen bond (H-bond) acidity, and ILs in the right side as the ILs with H-bond basicity. Indeed, [C_4mim][Cl] displays the highest H-bond basicity among the commonly used ILs as reported in ref. 10. KAT parameters for some synthetic polymers and natural polymers were determined by Harris *et al.*[23] and Spange *et al.*[24–27] Unfortunately, the KAT polarity parameters have not been employed systemically to analyze the polymer solubility in ILs yet.

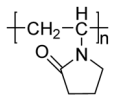
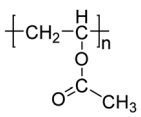
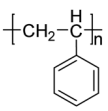
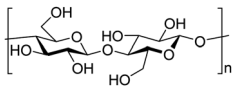
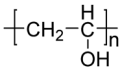
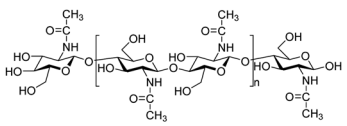
Rogers *et al.*[4] reported that the favorable dissolution of cellulose in [C_4mim][Cl] was mainly driven by H-bonded cross-association between the carbohydrate hydroxyl protons and the IL chloride ions. The solubility mechanism between celluloses (acidity) and ILs with H-bond basicity was validated by ^{13}C and $^{35/37}$Cl NMR.[28,29] The effect of hydrogen bond accepting ability (β) of anions on the dissolution of cellulose was considered further.[30,31] Recently, we realized that some polar synthetic polymers, which can be dissolved in ILs,[5,6] are the H-bonded basic polymers that normally have an electronegative atom. The ILs used in these solutions are ILs with H-bond acidity (*e.g.* [C_4mim][Tf_2N]) in the left side of Fig. 1. Therefore, it can be conjectured that the polymeric solubility in ILs may be dominated by the H-bonding interaction (acidity-basicity) where the difference in polarity between the solvent and solute is not so large.[10]

Results and discussion

Experimental observations of polymer solubility in ionic liquids

The contribution of H-bond on the polymeric solubility in ionic liquids is illustrated using typical H-bond basic and acidic polymers listed in Table 1.[23–27] Information of other polymers

Table 1 KAT parameters α, β, and π^* of typical polymers

Structure	Name	Abbreviation	α_p	β_p	π^*
	Poly(vinyl pyrrolidone)[a]	PVP	0.01 (−0.12)[b]	0.93 (0.78)[b]	0.93 (0.91)[b]
	Poly(vinyl acetate)[a]	PVAc	0.00 (0.02)[b]	0.40 (0.44)[b]	0.77 (0.76)[b]
	Polystyrene[b]	PS	0.08	0.06	0.65
	Microcrystalline cellulose[c]	MCC	1.31	0.62	0.34
	Poly(vinyl alcohol)[b]	PVA	0.66[d]	0.52	1.11
	Chitin[c]	—	0.67	0.86	0.91

[a] Cited from ref. 24. [b] Ref. 23. [c] Ref. 25. [d] α of PVA derived from model solvents.

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can be found in Tables S1 and S2 (ESI†). The structure and KAT parameters of 11 ILs[32,33] used in the present study are presented in Scheme 1 and Table 2. The subscripts after α and β refer to polymer (p), cation (+) or anion (−). Representative H-bonded basic polymers, poly(vinyl pyrrolidone) (PVP: α_P = 0.01, β_P = 0.93) and poly(vinyl acetate) (PVAc: 0.0, 0.40) usually contain H-bond accepting groups, such as –C–O–C–, or >C═O, or –COO–, which result in high β_P values, while the α_P values are zero or close to zero due to the absence of proton donor. Therefore, the H-bonded cross-association between the electronegative groups C═O and the hydrogen atoms of imidazole cations of the ILs with H-bond acidity (*e.g.* [C_4mim][Tf_2N]) promotes the solvation of polymers and subsequently the polymer solubility in ILs (see the solubility data in Table S3, ESI†). In contrast, the reference polymer, PS (0.08, 0.06) has neither HBA capacity nor HBD capacity, so no IL solvents can be found for PS (Table S3, ESI†). Moreover, the ILs with H-bond basicity, *e.g.* [C_4mim][Cl], do not dissolve PVP or PVAc (polymers with H-bond basicity), as expected from the H-bonding complementary principle. The solubility data of other polymers with H-bond basicity show that the high α_+ ILs always dissolve the high β_P polymers (Table S3, ESI†); hence, the "complementary" principle[34,35] of solubility (one component acidic and the other basic) is validated.

Representative polymers with H-bond acidity, microcrystalline celluloses (MCC: α_P = 1.31, β_P = 0.62), poly(vinyl alcohol) (PVA, 0.66, 0.52), and chitin (0.67, 0.86), contain acidic groups capable of H-bond formation, such as –OH, or –NH_2, or –COOH,[23–25] which result in high α_P values but the β_P values are high due to the presence of an electronegative atom (O or N). Therefore, the H-bonding self-association (polymer-polymer) hinders the dissolution of celluloses in common solvents. The ILs with strong H-bond basicity, *e.g.* [C_2mim][Ac] or [C_4mim][Cl], have β_- values higher than β_P, which overwhelm the H-bond basicity of polymers in favor of the formation of intermolecular H-bonding. Indeed, celluloses can be dissolved up to 25 wt% in [C_4mim][Cl][4] or [C_2mim][Ac].[36] In contrast, chitin has a β_P value higher than the β_- values of most ILs listed in Table 2, so only [C_2mim][Ac] (β_- = 0.99) can dissolve it. In contrast, ILs with H-bond acidity ([C_4mim][Tf_2N]) do not dissolve polymers with H-bond acidity (MCC or PVA), as expected by the H-bonding complementary principle. The solubility data of other polymers with H-bond acidity (Table S4, ESI†) support the conclusion that the basic ILs easily dissolve the acidic polymers.

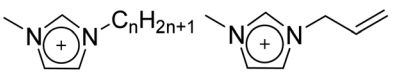
Table 2 KAT parameters α, β, and π^* for different ILs

	α_+		β_-		π^*	
Ionic liquids	Ref. 32	Ref. 33	Ref. 32	Ref. 33	Ref. 32	Ref. 33
[C_4mim][PF_6]	0.67	0.63	0.19	0.19	1.05	1.04
[C_4mim][Tf_2N]	0.69	0.64	0.25	0.25	0.97	0.97
[C_8mim][Tf_2N]	—	0.60	—	0.29	—	0.96
[C_8mim][BF_4]	—	0.62	—	0.41	—	0.98
[C_4mim][BF_4]	0.64	0.63	0.41	0.37	1.04	1.05
[C_8mim][PF_6]	0.58	—	0.46	—	0.88	—
[C_4mim][TfO]	0.62	0.62	0.46	0.49	1.03	1.00
[Amim][Br]	0.50	—	0.66	—	1.20	—
[C_4mim][Cl]	0.45	—	0.83	—	1.09	—
[Amim][Cl]	0.47	—	0.84	—	1.17	—
[C_2mim][Ac]	0.51	—	0.99	—	1.08	—

Dissolution mechanism

The dissolution mechanism of polymers with H-bond basicity and acidity in ILs are illustrated in Schemes 2a and b, respectively, where cross-association and self-association H-bonds have been considered. Strong cross-association and weak self-association favor solubility, while in solute/solvent pairs with weak cross-association and strong self-association, the mutual solubility is not favored.[35] Therefore, considering the competition between cross-association and self-association H-bonding, the quantitative hydrogen bonding (QHB) analysis is further proposed. A plot of $\Delta\alpha(\alpha_p - \alpha_+)$ *vs.* $\Delta\beta(\beta_P - \beta_-)$ is shown in Scheme 2c. Polymers with H-bond acidity, with $\Delta\alpha > 0$ and $\Delta\beta < 0$, favor the formation of cross-association H-bonding with basic ILs; hence, their solubility region should appear in the upper-left (second) quadrant of plane coordinates. Similarly, the solubility region of polymers with H-bond basicity in acidic ILs should appear in the lower-right (fourth) quadrant, where $\Delta\alpha < 0$ and $\Delta\beta > 0$. Moreover, greater the acidity and basicity of the intermolecular H-bonding atoms, stronger the

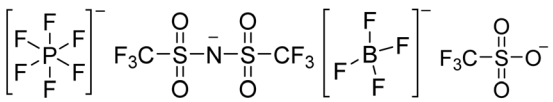
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[$C_n$mim]$^+$ [Amim]$^+$

1-alkyl-3-methylimidazolium 1-allyl-3-methylimidazolium

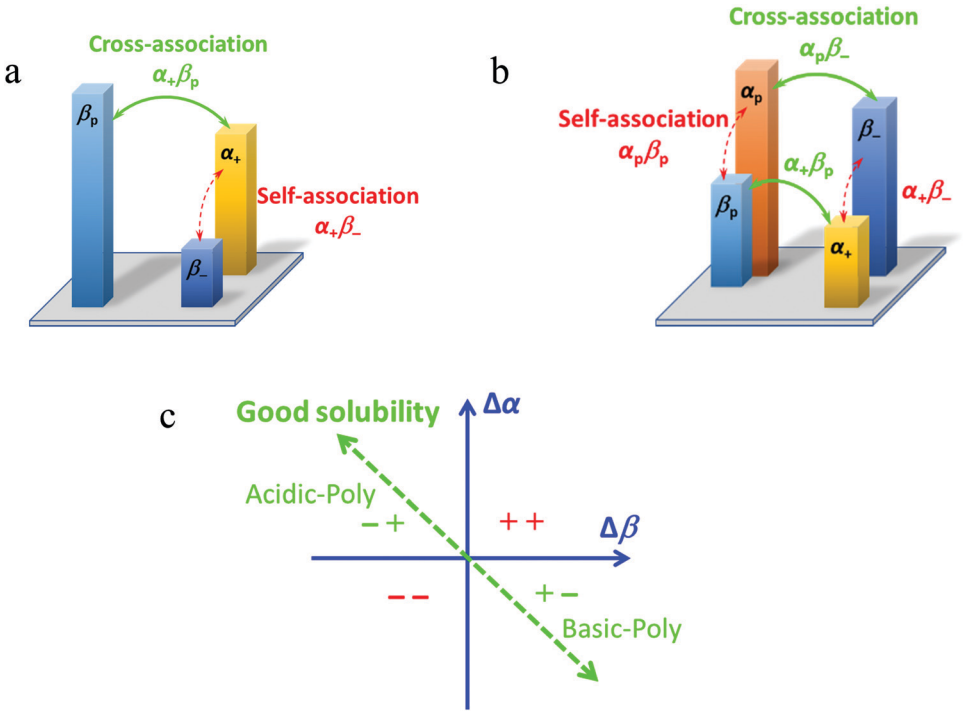
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PF_6^- [Tf_2N]$^-$ BF_4^- [TfO]$^-$

Scheme 1 Chemical structures and acronyms of the cations and anions of ILs.

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Scheme 2 Dissolution mechanism of a H-bond basicity polymer in acidic ILs (a), and a H-bond acidity polymer in basic ILs (b). (c) QHB analysis by a plot of $\Delta\alpha$ *vs.* $\Delta\beta$. Good solubility predicted in the second and fourth quadrants reflects the complementary nature of H-bond acidity and basicity scales.

H-bond would be, as indicated by the internal bisector crossing the second and fourth quadrants. Therefore, it is expected that the absolute value of the product of $\Delta\alpha\Delta\beta$ would be proportional to the solubility of polymers in ILs (green arrow direction, Scheme 2c).

According to the QHB analysis, the solubility data of PVP, PVAc and PS in ILs are presented in Fig. 2a. Clearly, the solubility region appears in the fourth quadrant as predicted (Scheme 2c), and the solubility order is PVP ($\beta_P = 0.93$) > PVAc (0.40) > PS (0.06), as discussed above. In this case, the influence of α_p ($\sim$0) is negligible. On the other hand, the solubility region of polymers with H-bond acidity appears in the second quadrant (Fig. 2b) according to the QHB analysis (Scheme 2c). However, the contributions of β_P and α_+, as the secondary cross H-bond, need to be accounted.[36]

It is interesting to observe that PVA is soluble in ILs with H-bond basicity and PVAc in ILs with H-bond acidity. Indeed, PVA is prepared by the hydrolysis of PVAc as the ester group is transformed into the hydroxy group (Table 1). The consequent change in the solubility from PVAc to PVA suggests that the polymeric solubility in ILs is certainly dominated by H-bonding. On the other hand, polymers that are neither very basic nor very acidic (*e.g.*, PS or other polyolefins) cannot be dissolved in ILs. This observation also supports the dissolution mechanism from another aspect.

Following the solubility criteria dominated by H-bonding interactions, the solubility of other polar polymers was also tested (see Tables S3 and S4, ESI,† in which the product of $\Delta\alpha\Delta\beta \times 10^3$ are listed.). The QHB analysis for all the polymers is presented in Fig. 2c. The solubility criteria illustrated in Scheme 2c have been validated by the solubility data of 19 polymers (11 acidic, and 8 basic) in 11 ILs (7 acidic, and 4 basic). It should be noted that some data closed to the boundary (the axis) are not very consistent with others since the uncertainty of KAT parameters reported in the literature is 0.1 at least.[23,24,32,33]

Three principles of hydrogen-bonding interactions

The failure of solubility parameters in polymer-IL solutions asked for a better understanding of how to handle H-bonding interactions. H-bond can be considered as an electrostatic interaction,[37] and two atoms forming H-bond are electronically complementary (or acidity-basicity complementary), although the directional preferences should be further considered.[38] According to the "like dissolves like" principle,[12] ΔH_m, the enthalpy change on mixing is proportional to $(\delta_1 - \delta_2)^2$, which must be small for the components to be miscible. Hansen[13] extended the Hildebrand parameter method to polar and hydrogen-bonding systems, and Hansen's solubility parameters treat the dispersion (d), polar (p), and H-bonding (HB) contributions in the same way:

$$\Delta H_m = \chi\phi_p\phi_s RT \sim (\delta_{d_p} - \delta_{d_s})^2 + (\delta_{p_p} - \delta_{p_s})^2 + (\delta_{HB_p} - \delta_{HB_s})^2, \quad (3)$$

Therefore, the H-bonding contribution towards the affinity of a solute/solvent pair is higher when the values of their H-bonding

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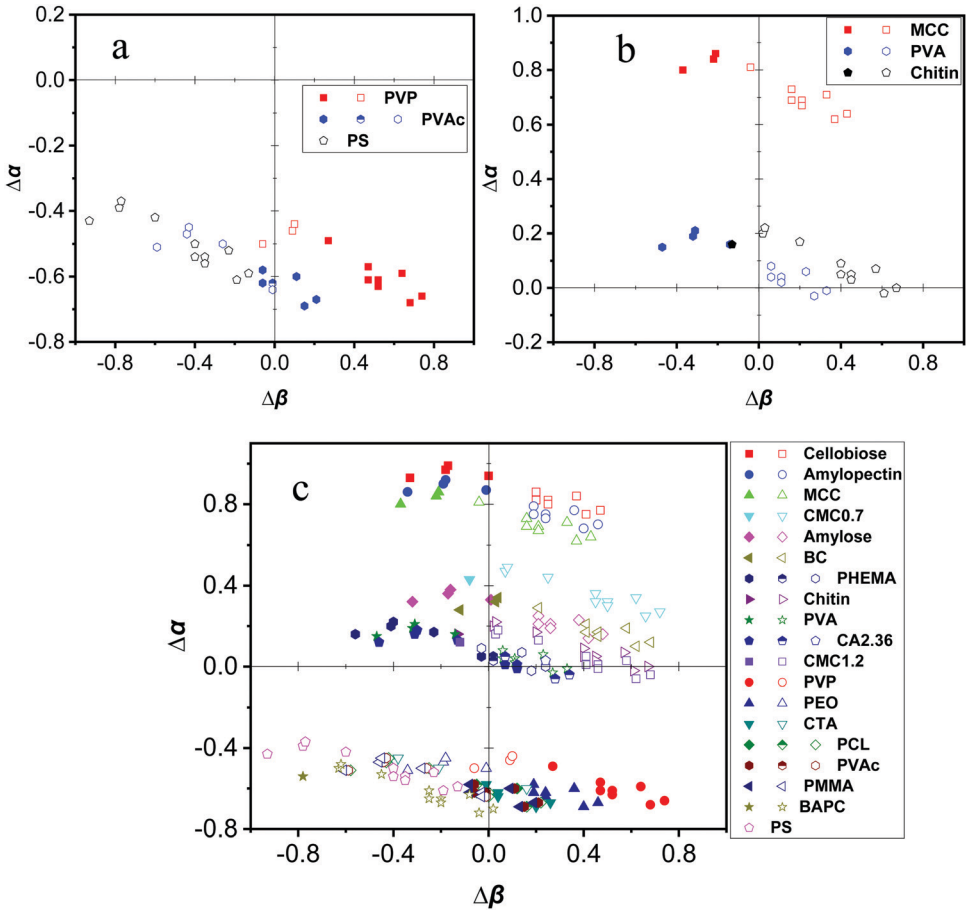


Fig. 2 QHB analysis of representative H-bond basicity polymers (a), H-bond acidity polymers (b), and all polymers tested in this study (c). Solid symbols: solubilization; half-filled symbols: solubilization at high temperatures only; open symbols: insolubilization.

SP component are closer, which can be indicated as the coordinate origin in Scheme 2c. Clearly, the SP approach for the H-bonding is inadequate since experiments indicate that the H-bonding interactions promote the solubility, and also validate that the solubility is related to the product of $\Delta\alpha\Delta\beta$ in Fig. 2c. A rational handling of the H-bonding contribution based on KAT parameters is proposed, including features as follow:

First, the hydrogen-bonding contribution to the affinity of a solute/solvent pair is obtained from the complementary product. Indeed, the product of an acidity (proton donor) with a basicity (proton acceptor) parameter, $\alpha\beta$, reflects the electronical complementary nature of H-bond acidity and basicity scales, and that is, between a proton donor with partial positive charge and a proton acceptor with partial negative charge.[34,35,37,38]

Second, the competition between cross-association and self-association H bonding is considered as the difference between the H-bonding acidity of polymer and IL, $\Delta\alpha$ ($\alpha_p - \alpha_+$) and the basicity difference, $\Delta\beta$ ($\beta_P - \beta_-$). Higher the absolute value of the product $\Delta\alpha\Delta\beta$, higher is the cross-association H-bond strength, and higher is the solubility (Scheme 2c and Fig. 2c).

Third, the QHB analysis reflects the free-energy nature[35] of the H-bonding interactions: $\Delta G_{HB} = \sim\Delta\alpha\Delta\beta$. Indeed, by mathematically unfolding this product, we obtained $-(\alpha_p\beta_- + \alpha_+\beta_P) + (\alpha_p\beta_P + \alpha_+\beta_-)$. The first sum, which represents that cross-associations *favor* solubility, is *negative*. The second sum is *positive* as self-associations hinder dissolution. Therefore, the solubility regions appear in the second and fourth quadrants in which the sign of coordinates are opposite (Scheme 2c).

In summary, for the polymer–IL mixture, the Gibbs free energy of mixing is expressed as the contribution of H-bonding interactions is considered properly according to three principles of H-bonding interactions:

$$\begin{aligned}\Delta G_m &= \Delta H_m - T\Delta S_m \\ &= RT\left(\frac{\phi_p}{N_p}\ln\phi_p + \phi_{IL}\ln\phi_{IL} + \chi\phi_p\phi_{IL} + c_{HB}\Delta\alpha\Delta\beta\right),\end{aligned} \tag{4}$$

and c_{HB} is a constant related to the H-bonding strength. These items in the right side of eqn (4) represent the contributions of the entropy of mixing, enthalpy of mixing from non H-bonding interactions and H-bonding interactions, respectively. Clearly, the treatment for the contribution of H-bonding interactions is totally different from the Hanson approach (eqn (3)): $\Delta\alpha\Delta\beta$ *vs.* $(\delta_{HB_p} - \delta_{HB_s})^2$.

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Actually, the KAT H-bond acidity parameter α is determined by means of the solvatochromic dyes with a H-bond basicity group, while the parameter β is determined by the dyes with an acidic group.[18] The H-bonding complementary principle has been employed; hence, the KAT parameters reflect the molecular details of H-bonding interactions. In addition, the scales of H-bond acidity and basicity of ionic liquids have been validated by COSMO-RS calculations,[39,40] which is helpful to understand the polymer solubility in ionic liquids.[41,42] It is worth noting that the thermodynamic models also support the H-bonding interaction mechanism between ionic fluids and polymers or biomolecules.[43–45]

Conclusions

Conclusively, the polymer solubility in ILs is rationalized by properly considering the H-bonding interactions. The QHB analysis will serve as a simple pre-screening tool for the determination of polymer solubility in ILs. For numerous other systems strongly influenced by H-bonding interactions, a better understanding should be achieved by the QHB analysis. For example, as a novel green solvent, deep eutectic solvents, which are mixtures of two components capable of interacting *via* hydrogen bonds to form a eutectic mixture, exhibit excellent applications in polymer chemistry,[46] natural product extraction,[47] removal of volatile organochlorine compounds.[48] Solid amorphous dispersions technology[49] improves the solubility of poorly-water soluble drug molecules (with many H-bonding sites) by miscible blending with polymeric carriers, including PVP, PEG, cellulose derivations discussed in this study. Cationized cellulose esters/acidity IL composite membranes exhibit high-performance CO_2 (basicity) separation,[50] and CO_2 electroreduction in ionic liquid-based electrolytes demonstrate high CO_2 adsorption capacity, high selectivity, and low energy consumption.[51] The introduction of functionalized ILs into polymer networks can realize the preparation of mechanically robust smart ionogels.[52] It has been realized that these systems are dominated by hydrogen-bonding interactions; hence, principles presented in this study will be very helpful to optimize these processes.

Conflicts of interest

There are no conflicts of interest to declare.

Acknowledgements

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